

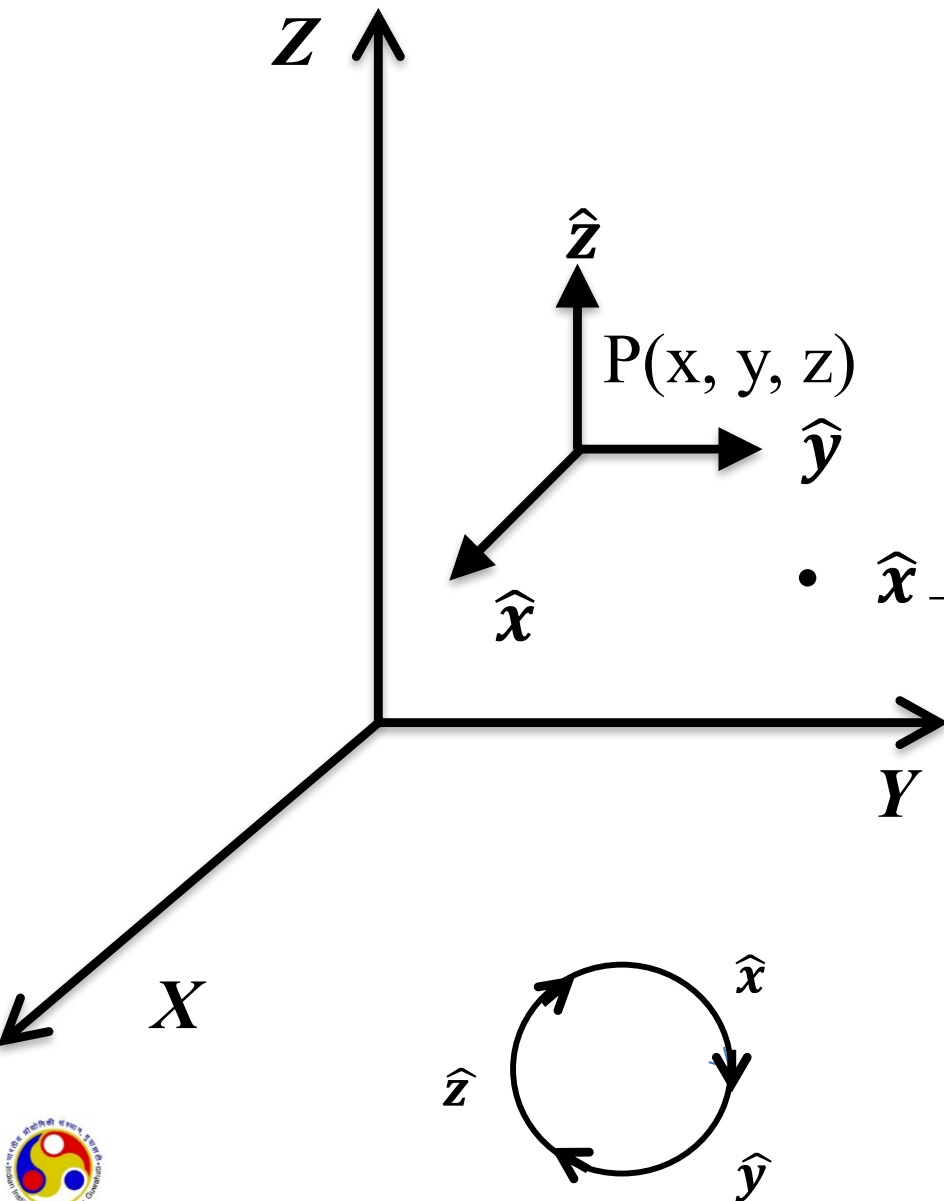
Lecture 04

Coordinate systems

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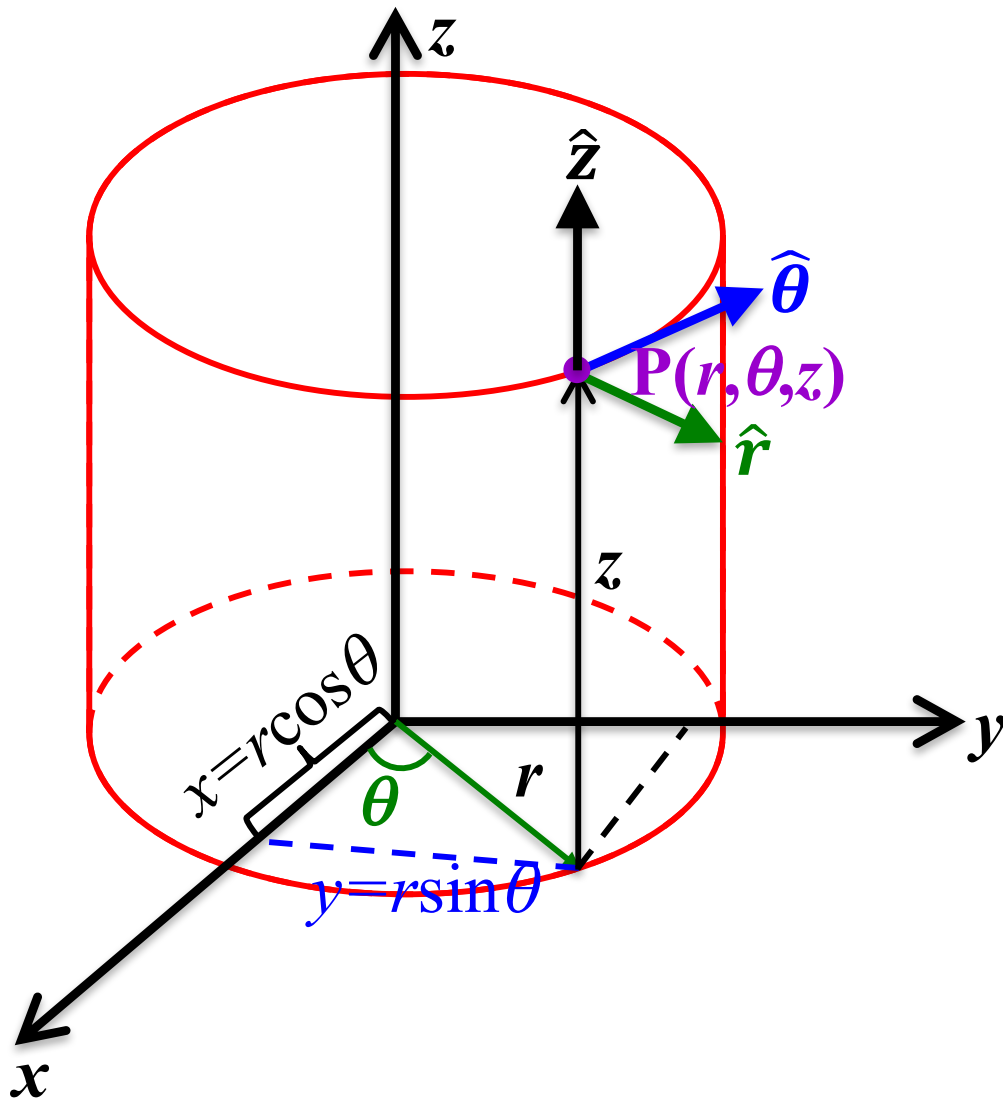
Cartesian co-ordinate system in 3D



- In Cartesian coordinate system, one can define three unit vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ at every point.
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors in the increasing direction of x, y and z.
- $\hat{x} \perp$ to the YZ plane $\hat{y} \perp$ to the XZ plane and $\hat{z} \perp$ to the XY plane
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are orthogonal, and directed in the same direction at every points.

$$\hat{x} \times \hat{y} = \hat{z}; \hat{y} \times \hat{z} = \hat{x}; \hat{z} \times \hat{x} = \hat{y};$$

Cylindrical Coordinate System



How to locate a point 'P' in space ?

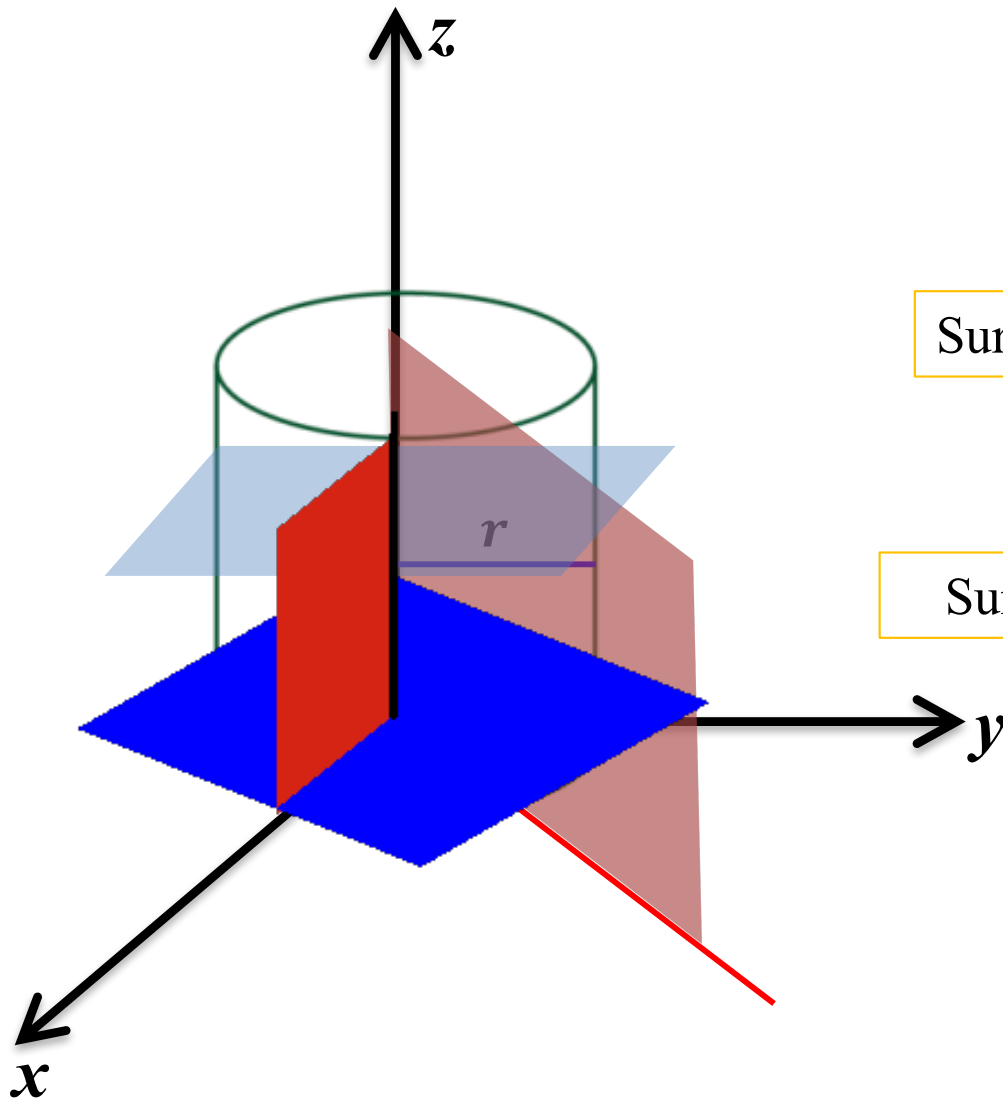
(r, θ, z) coordinates system is known as cylindrical coordinate system

(r, θ) Coordinate of the foot of the point in XY plane

z-Height from the XY plane

$\hat{r}$, $\hat{\theta}$, and $\hat{z}$ are unit vectors along increasing direction of r , θ , and z .

Cylindrical Coordinate System



Note that point (r, θ, z) is at the intersection of three surfaces:

The surface with $r = \text{Constant}$; Cylinder

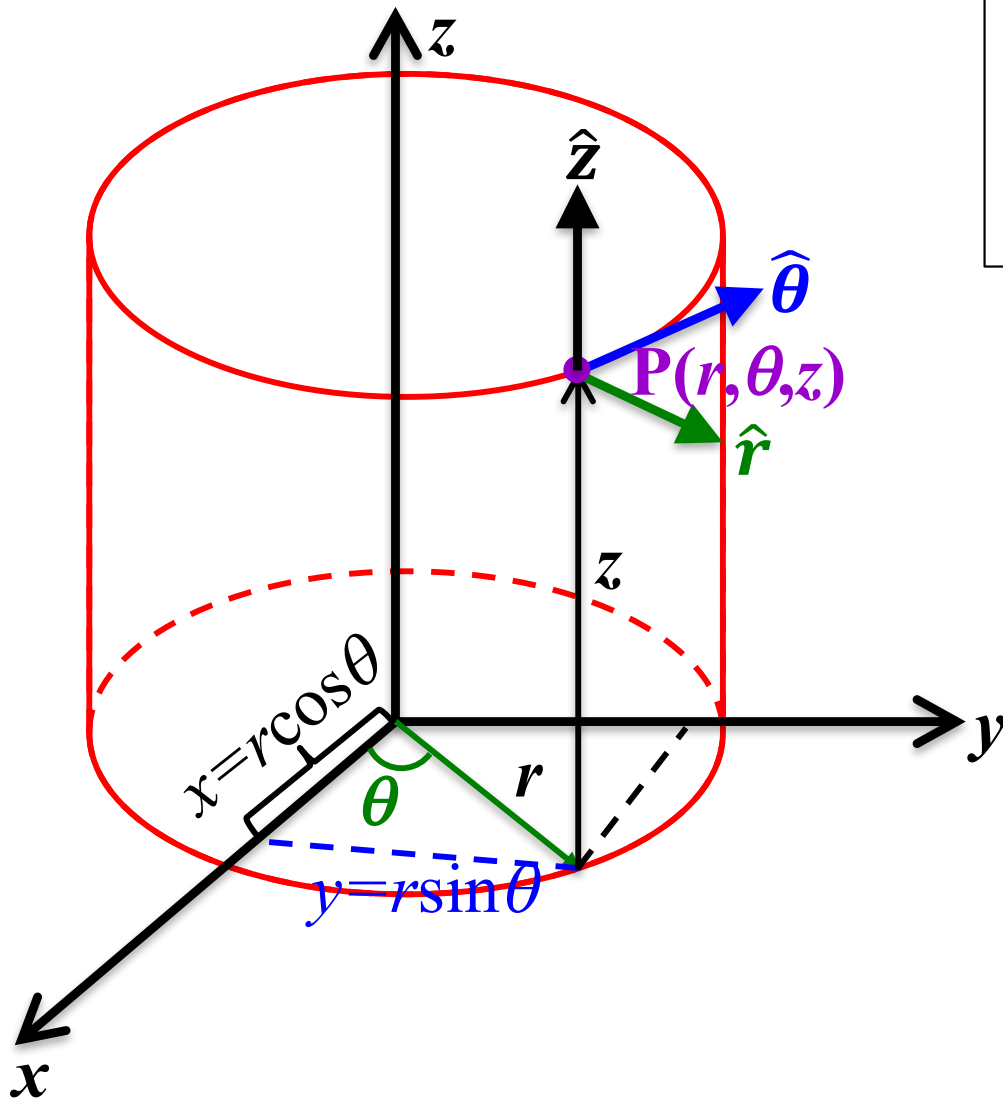
Surface with $\theta = \text{Constant}$; Plane containing Z-axis

Surface $z = \text{Constant}$. Plane Perpendicular Z-axis

The three orthogonal components, r (green), θ (red), and z (blue), each increasing at a constant rate. The point is at the intersection between the three colored surfaces.



Cylindrical Coordinate System



Transformation equation is very similar to polar coordinate with additional z-coordinate.

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta \\z &= z\end{aligned}$$

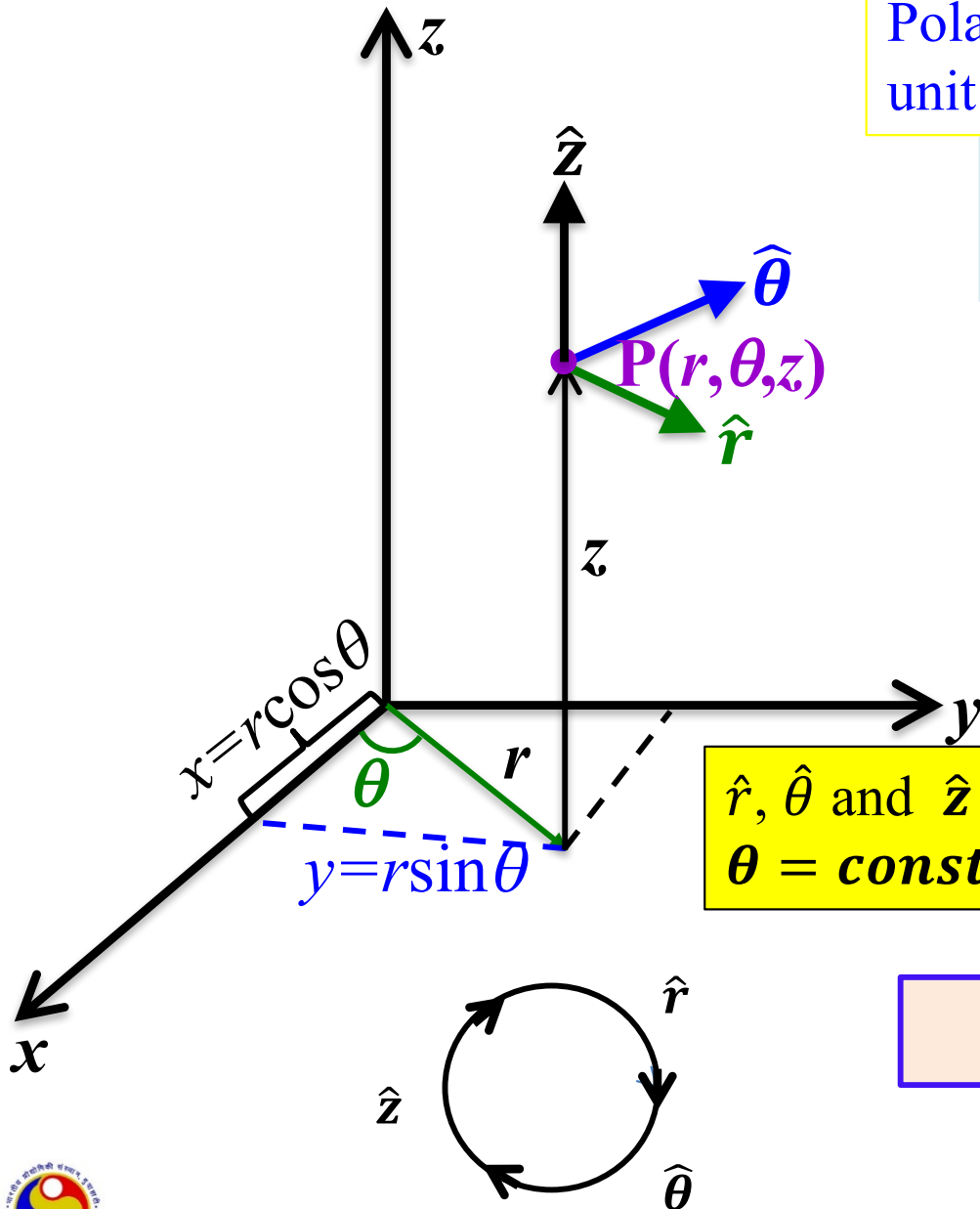
Reverse transformation

$$\begin{aligned}r &= (x^2 + y^2)^{1/2} \\ \theta &= \tan^{-1} \frac{y}{x} \\ z &= z\end{aligned}$$

Note: Instead of (r, θ) , many books use notation (ρ, φ) .



Cylindrical Coordinate System



Polar coordinate unit vectors $(\hat{r}, \hat{\theta})$ + additional unit vector in the z -direction.

$\hat{r}$, $\hat{\theta}$ and $\hat{z}$ are unit vectors along increasing direction of coordinates r , θ and z .

$$\begin{aligned}\hat{r} &= \cos \theta \hat{x} + \sin \theta \hat{y} \\ \hat{\theta} &= -\sin \theta \hat{x} + \cos \theta \hat{y} \\ \hat{z} &= \hat{z}\end{aligned}$$

$\hat{r}$, $\hat{\theta}$, and $\hat{z}$ are **orthogonal**, the directions of $\hat{r}$, $\hat{\theta}$, depend on location but $\hat{z}$ is fixed

$\hat{r}$, $\hat{\theta}$ and $\hat{z}$ are **perpendicular** to surfaces $r = \text{constant}$; $\theta = \text{constant}$ and $z = \text{constant}$, respectively.

$$\hat{r} \times \hat{\theta} = \hat{z}; \quad \hat{\theta} \times \hat{z} = \hat{r}; \quad \hat{z} \times \hat{r} = \hat{\theta}$$



Position, Velocity, Acceleration, Newton's law in cylindrical coordinate system

Vector components are very similar to polar coordinate + z component

Position vector

$$\vec{r} = r\hat{r} + z\hat{z}$$

Velocity

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{z}$$

$$\frac{d\hat{z}}{dt} = 0$$

Acceleration

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}$$

Newton's law

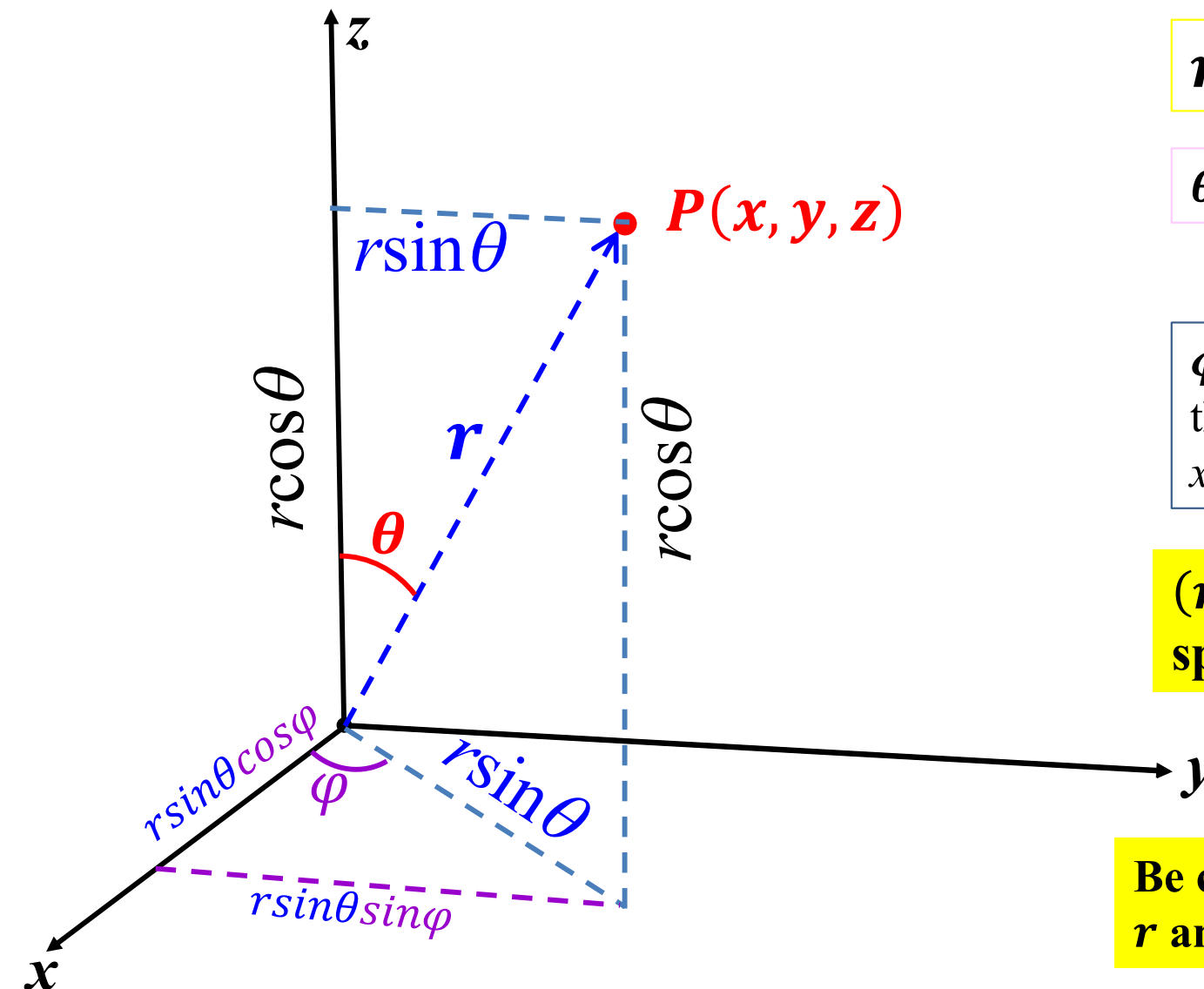
$$\begin{aligned}\vec{F} &= F_r\hat{r} + F_\theta\hat{\theta} + F_z\hat{z} \\ &= m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}]\end{aligned}$$



Spherical polar coordinate



Spherical polar coordinate system



$r \rightarrow$ Radial distance from origin

$\theta \rightarrow$ Angle with z -axis.

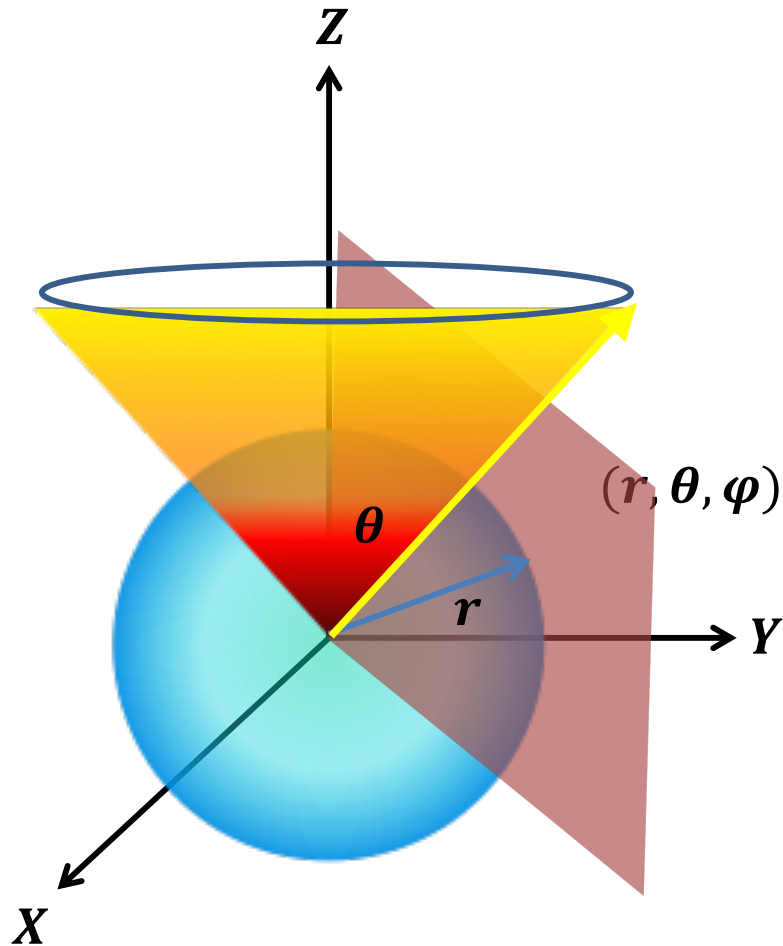
$\varphi \rightarrow$ Angle between x -axis and the projection of radial vector in xy plane

(r, θ, φ) is known as spherical polar coordinate

Be careful, notations are different.
 r and θ are not planer coordinate.



Spherical polar coordinate system



Note that point (r, θ, ϕ) is at the intersection of three surfaces:

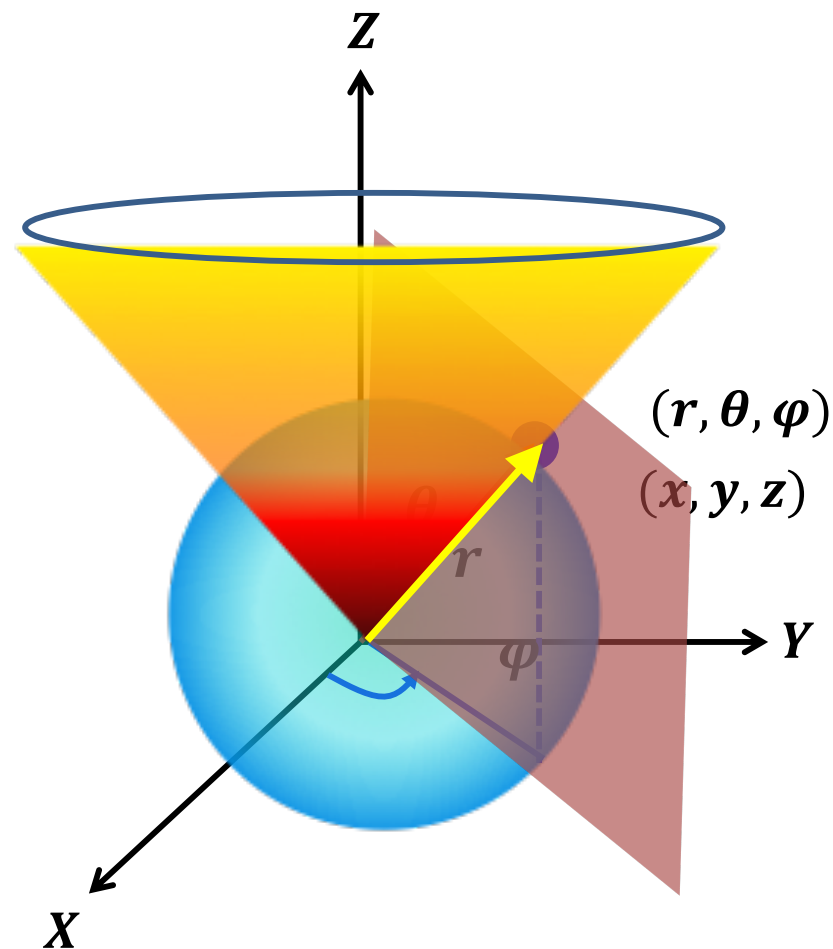
The surface with $r = \text{Constant}$; Sphere

Surface with $\theta = \text{Constant}$; Cone

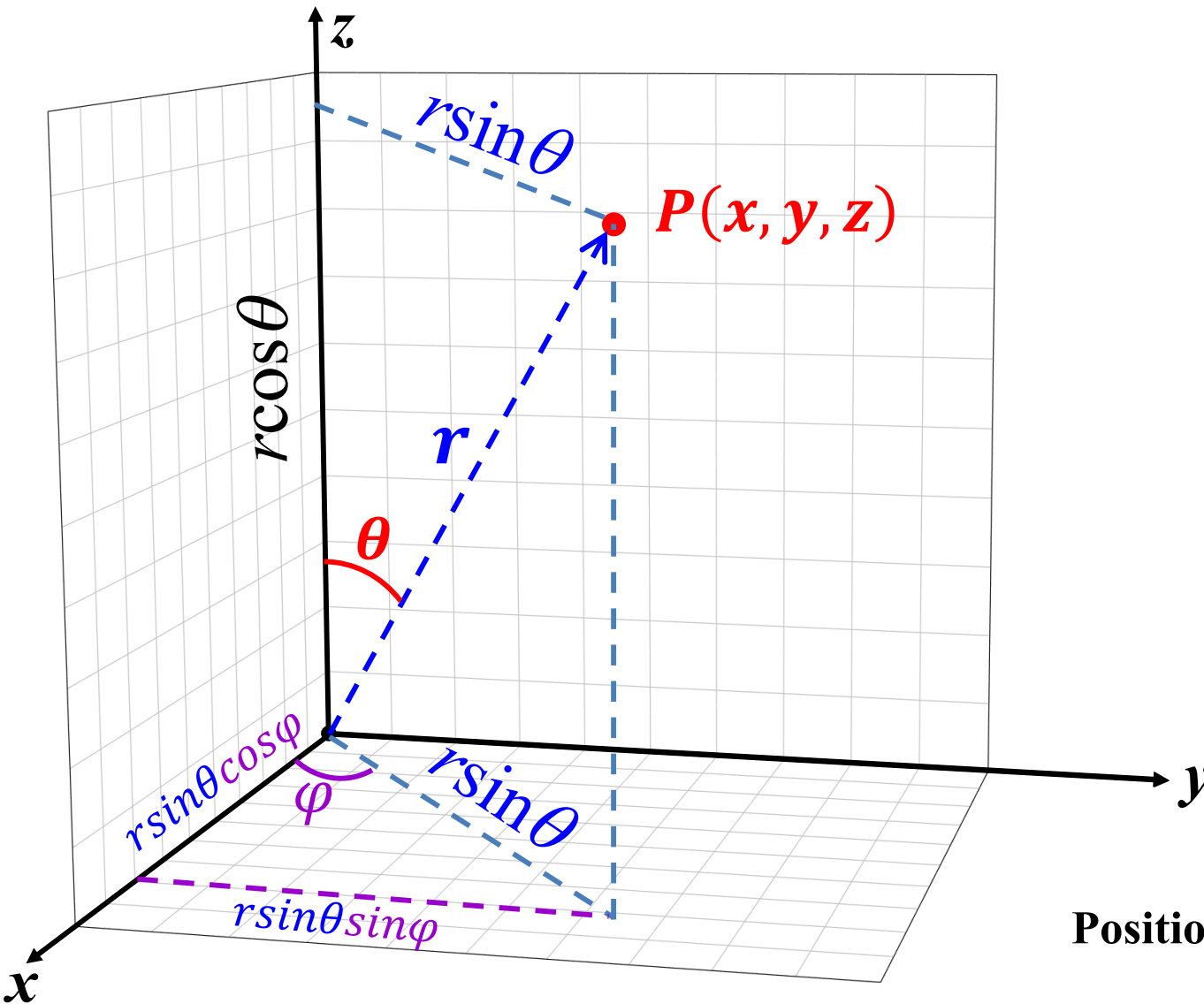
Surface $\phi = \text{Constant}$. Plane containing Z-axis

Spherical polar coordinate system

So the point (r, θ, φ) is at the intersection of three surfaces:



Spherical polar coordinate system



Transformation relations

$$x = r \sin \theta \cos \varphi$$

$$y = r \sin \theta \sin \varphi$$

$$z = r \cos \theta$$

Hence

$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$\theta = \tan^{-1} \left(\frac{(x^2 + y^2)^{1/2}}{z} \right)$$

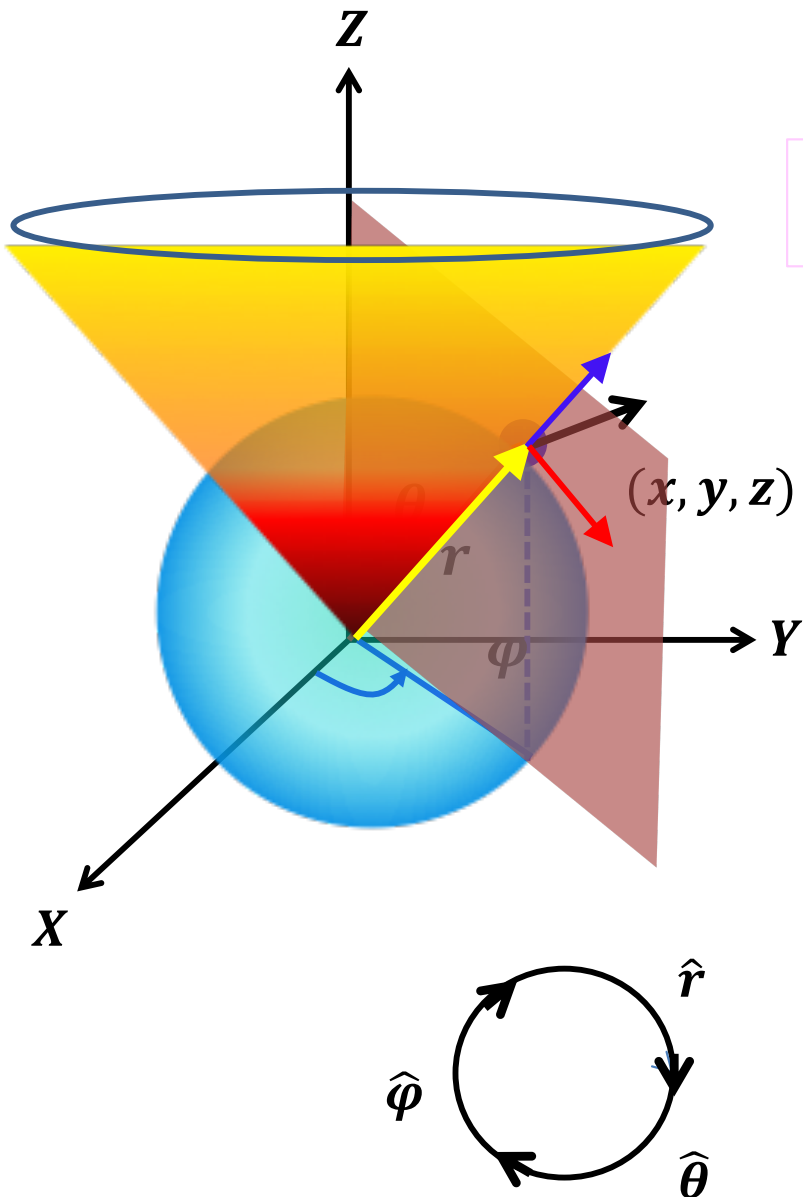
$$\varphi = \tan^{-1} \left(\frac{y}{x} \right)$$

Position vector in terms of x , y and z components

$$\vec{r} = r \sin \theta \cos \varphi \hat{x} + r \sin \theta \sin \varphi \hat{y} + r \cos \theta \hat{z}$$



Unit vectors: Spherical polar coordinate system



Position vector

$$\vec{r} = r \sin \theta \cos \varphi \hat{x} + r \sin \theta \sin \varphi \hat{y} + r \cos \theta \hat{z}$$

Unit vectors: $\hat{r} = \left(\frac{\partial \vec{r} / \partial r}{|\partial \vec{r} / \partial r|} \right)$, $\hat{\theta} = \left(\frac{\partial \vec{r} / \partial \theta}{|\partial \vec{r} / \partial \theta|} \right)$, $\hat{\varphi} = \left(\frac{\partial \vec{r} / \partial \varphi}{|\partial \vec{r} / \partial \varphi|} \right)$

$$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$\hat{r}$, $\hat{\theta}$, and $\hat{\varphi}$ are perpendicular to $r = \text{constant}$,
 $\theta = \text{constant}$, $\varphi = \text{constant}$.

We can also get the unit vectors

$$\hat{r} = \hat{\theta} \times \hat{\varphi}; \quad \hat{\theta} = \hat{\varphi} \times \hat{r}; \quad \hat{\varphi} = \hat{r} \times \hat{\theta}$$

Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of θ and φ only.

Thus,

$$\frac{\partial \hat{r}}{\partial r} = 0$$

$$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$$\begin{aligned} \frac{\partial \hat{r}}{\partial \theta} &= \frac{\partial}{\partial \theta} (\sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}) \\ \frac{\partial \hat{r}}{\partial \theta} &= \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z} = \hat{\theta} \end{aligned}$$

$$\begin{aligned} \frac{\partial \hat{r}}{\partial \varphi} &= \frac{\partial}{\partial \varphi} (\sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}) \\ \frac{\partial \hat{r}}{\partial \varphi} &= -\sin \theta \sin \varphi \hat{x} + \sin \theta \cos \varphi \hat{y} + 0 = \sin \theta \hat{\varphi} \end{aligned}$$

$$\frac{\partial \hat{\theta}}{\partial r} = 0$$

$$\begin{aligned} \frac{\partial \hat{\theta}}{\partial \theta} &= \frac{\partial}{\partial \theta} (\cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}) \\ \frac{\partial \hat{\theta}}{\partial \theta} &= (-\sin \theta \cos \varphi \hat{x} - \sin \theta \sin \varphi \hat{y} - \cos \theta \hat{z}) = -\hat{r} \end{aligned}$$



Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of θ and φ only.

$$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$$\frac{\partial \hat{\theta}}{\partial \varphi} = \frac{\partial}{\partial \varphi} (\cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \hat{z} \sin \theta)$$

$$\frac{\partial \hat{\theta}}{\partial \varphi} = (-\cos \theta \sin \varphi \hat{x} + \cos \theta \cos \varphi \hat{y} - 0) = \cos \theta \hat{\varphi}$$

$$\frac{\partial \hat{\varphi}}{\partial r} = 0$$

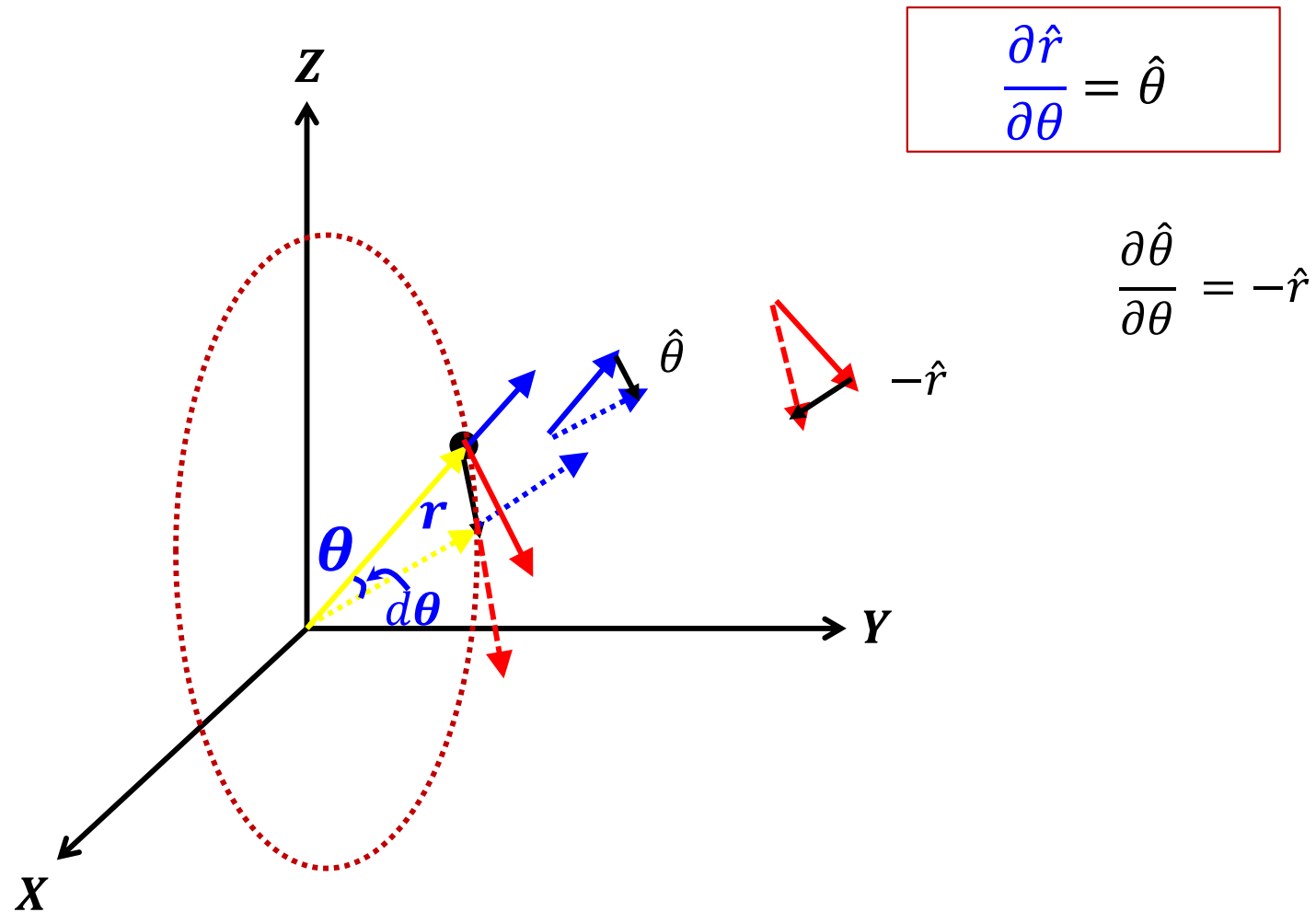
$$\frac{\partial \hat{\varphi}}{\partial \theta} = \frac{\partial}{\partial \theta} (-\sin \varphi \hat{x} + \cos \varphi \hat{y}) = 0$$

$$\frac{\partial \hat{\varphi}}{\partial \varphi} = \frac{\partial}{\partial \varphi} (-\sin \varphi \hat{x} + \cos \varphi \hat{y})$$

$$\frac{\partial \hat{\varphi}}{\partial \varphi} = -\cos \varphi \hat{x} - \sin \varphi \hat{y} = -(\cos \varphi \hat{x} + \sin \varphi \hat{y})$$

$$\frac{\partial \hat{\varphi}}{\partial \varphi} = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta)$$





Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of θ and φ only.

$$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

$$\frac{\partial \hat{r}}{\partial r} = 0$$

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$$

$$\frac{\partial \hat{r}}{\partial \varphi} = \sin \theta \hat{\varphi}$$

$$\frac{\partial \hat{\theta}}{\partial r} = 0$$

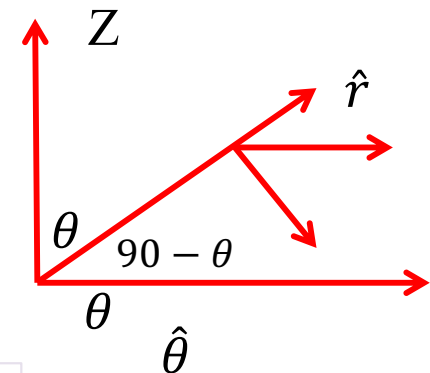
$$\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

$$\frac{\partial \hat{\theta}}{\partial \varphi} = \cos \theta \hat{\varphi}$$

$$\frac{\partial \hat{\varphi}}{\partial r} = 0$$

$$\frac{\partial \hat{\varphi}}{\partial \theta} = 0$$

$$\frac{\partial \hat{\varphi}}{\partial \varphi} = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta)$$



A line in XY plane at angle φ with x-axis

$$-\sin \varphi \hat{x} + \cos \varphi \hat{y} = \hat{\varphi}$$

$$\cos \varphi \hat{x} + \sin \varphi \hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$

$$\hat{x} = \sin \theta \cos \varphi \hat{r} + \cos \theta \cos \varphi \hat{\theta} - \sin \varphi \hat{\varphi}$$

$$\hat{y} = \sin \theta \sin \varphi \hat{r} + \cos \theta \sin \varphi \hat{\theta} + \cos \varphi \hat{\varphi}$$

$$\hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$



Velocity in Spherical coordinate

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{r})$$

$$= \frac{dr}{dt}\hat{r} + r \frac{d\hat{r}}{dt}$$

$$= \dot{r}\hat{r} + r \left(\frac{\partial \hat{r}}{\partial \theta} \frac{d\theta}{dt} + \frac{\partial \hat{r}}{\partial \varphi} \frac{d\varphi}{dt} \right) \quad \leftarrow \text{Chain rule}$$

$$\vec{v} = \dot{r}\hat{r} + r(\dot{\theta}\hat{\theta} + \sin\theta\dot{\varphi}\hat{\varphi})$$

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\dot{\varphi}\hat{\varphi}$$

$$\hat{r} = \sin\theta\cos\varphi\hat{x} + \sin\theta\sin\varphi\hat{y} + \cos\theta\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\hat{x} + \cos\theta\sin\varphi\hat{y} - \sin\theta\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\hat{x} + \cos\varphi\hat{y}$$

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta} \quad \frac{\partial \hat{r}}{\partial \varphi} = \sin\theta\hat{\varphi}$$

You need to try this out?

Can you obtain Acceleration??

$$\begin{aligned} \vec{a} &= (\ddot{r} - r\dot{\theta}^2 - r\dot{\varphi}^2\sin^2\theta)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &\quad - r\dot{\varphi}^2\sin\theta\cos\theta)\hat{\theta} + (r\ddot{\varphi}\sin\theta + 2r\dot{\theta}\dot{\varphi}\cos\theta + 2\dot{r}\dot{\varphi}\sin\theta)\hat{\varphi} \end{aligned}$$



Velocity in Spherical coordinate

$$\begin{aligned}\vec{a} &= (\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2 \sin^2 \theta) \hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} \\ &\quad - r\dot{\phi}^2 \sin \theta \cos \theta) \hat{\theta} + (r\ddot{\phi} \sin \theta + 2r\dot{\theta}\dot{\phi} \cos \theta + 2\dot{r}\dot{\phi} \sin \theta) \hat{\phi}\end{aligned}$$

$$a_r = (\ddot{r} - r\dot{\theta}^2 - r\sin\theta\dot{\phi}^2\sin\theta)$$

Centripetal part due to angular motion
in a plane containing Z- axis

Centripetal part due to angular
motion in xy plane

$$a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\dot{\phi}^2\cos\theta)$$

Coriolis part due to angular motion
in a plane containing Z- axis

Centripetal part due to angular
motion in xy plane

$$a_\phi = (r\ddot{\phi} \sin \theta + 2r\dot{\theta}\cos\theta\dot{\phi} + 2\dot{r}\sin\theta\dot{\phi})$$

Coriolis part due to angular
motion in xy plane

